

AMAL P K

Cloud & DevOps Engineer

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PROFESSIONAL SUMMARY

Cloud and DevOps Engineer with hands-on experience building and automating production-grade infrastructure on AWS. Proficient in CI/CD pipeline design, Infrastructure as Code (Terraform, Ansible), container orchestration with Kubernetes, and observability with Prometheus and Grafana. Complemented by a background in Python and Node.js backend development, enabling end-to-end ownership from code to deployment.

TECHNICAL SKILLS

Cloud & Infra	AWS — EC2, EKS, ECS, S3, RDS, ALB, IAM, VPC, ACM, CloudWatch
Containers	Docker, Docker Compose, Kubernetes, Helm, Amazon EKS
CI/CD & IaC	GitHub Actions, Jenkins, Terraform, Ansible, Shell Scripting
Monitoring	Prometheus, Grafana, Alertmanager, Nginx, ALB
Backend	Python, Node.js, RESTful API Design, Express.js
Databases	PostgreSQL, MySQL, MongoDB
Tools & Platforms	Git, GitHub, Linux, Postman

WORK EXPERIENCE

AWS DevOps Engineer Trainee | Cokonet Academy, Trivandrum

Dec 2025 – Present

- Provisioned and maintained cloud infrastructure on AWS (EC2, S3, IAM, VPC, EKS) following security best practices — least-privilege IAM policies and private-subnet isolation.
- Built multi-stage CI/CD pipelines with Jenkins and GitHub Actions, eliminating manual deployment steps and reducing release friction across environments.
- Containerised applications with Docker and orchestrated deployments on Amazon EKS, ensuring rolling updates and zero-downtime releases.
- Automated infrastructure provisioning using Terraform and Ansible, enabling repeatable, version-controlled environment creation.
- Configured Nginx as a reverse proxy and load balancer for containerised workloads in production environments.

Junior Backend Developer | Adviciya Solutions, Ernakulam

Jun 2025 – Dec 2025

- Designed and developed RESTful APIs in node.js, nest.js implementing core business logic and optimising PostgreSQL queries for improved response times.
- Integrated GCP Cloud Tasks for distributed background job scheduling and Firebase Cloud Messaging (FCM) for push notifications.
- Managed IAM roles, VPC configurations, and network security rules across cloud environments.
- Worked in Git-based workflows with feature branching, pull-request code reviews, and structured release management.

Full Stack Web Development Intern | Edunet Foundation (EY GDS & AICTE), Remote

Jan 2025 – Mar 2025

- Completed a 6-week MERN stack internship covering full-stack application architecture and deployment.
- Built a personal finance tracker with CRUD operations, JWT/bcrypt authentication, and Chart.js dashboards.
- Developed and integrated RESTful APIs connecting a React frontend to a Node.js/Express backend.

EDUCATION

B.Tech — Information Technology · APJ Abdul Kalam Technological University, Thalassery

CGPA 7.5

Diploma — Wood & Paper Technology · Govt Polytechnic College, Kannur

CGPA 8.6

Higher Secondary — Biology Science · Kerala Board

70%

PROJECTS

Dockerized MERN Application [GitHub ↗](#)

- Containerised a full-stack MERN app with Docker Compose, orchestrating frontend, backend, and MongoDB with health-check-gated startup.
- Built a multi-stage Docker image for the React frontend (Node Alpine build → Nginx Alpine runtime), keeping the build toolchain out of the final image.
- Configured Nginx as a reverse proxy, serving static assets and routing API requests through a single entry point.

Tech: *Docker, Docker Compose, Nginx, React, Node.js, Express.js, MongoDB, AWS EC2*

CI/CD Pipeline — Spring Boot on GitHub Actions, Docker & EC2 [GitHub ↗](#)

- Built a GitHub Actions pipeline that runs the Maven test suite, builds a multi-stage Docker image, and pushes to Amazon ECR on every push to main — tagged with both Git commit SHA and latest.
- Replaced static AWS access keys with IAM OIDC federation, enabling GitHub Actions to assume a scoped IAM role with short-lived credentials.
- Automated EC2 deployment over SSH: pulls the new image, replaces the running container, and verifies the rollout via a /health check before marking the job successful.
- Reduced production image size by separating build and runtime stages — compiling with the full Maven/JDK toolchain, then shipping only the JAR on a minimal JRE-Alpine base.

Tech: *Java, Spring Boot, Maven, GitHub Actions, Docker, Amazon ECR, Amazon EC2, AWS IAM OIDC*

Multi-Tier AWS Infrastructure with Terraform [GitHub ↗](#)

- Designed and provisioned a multi-tier AWS architecture (VPC, public EC2 subnet, private multi-AZ RDS subnets, security groups, IAM roles) using Terraform with remote S3 state, isolating the database tier from direct internet access.
- Automated EC2 provisioning and FastAPI deployment via idempotent Ansible roles — installing Docker, building the container image, and orchestrating it with Docker Compose.
- Eliminated hardcoded credentials by storing DB connection parameters in AWS SSM Parameter Store (SecureString) and granting access exclusively through an EC2 IAM instance role.

Tech: *Terraform, Ansible, AWS VPC, EC2, RDS PostgreSQL, S3 Backend, IAM, SSM Parameter Store, Docker, FastAPI.*

Amazon EKS Deployment [GitHub ↗](#)

- Deployed a containerised application on Amazon EKS with node groups, namespaces, Deployments, Services, and Ingress resources.
- Set up AWS Load Balancer Controller and ALB Ingress for external traffic with SSL termination via ACM.
- Implemented Horizontal Pod Autoscaler (HPA) to scale pods on CPU utilisation for high availability.

Tech: *Amazon EKS, Kubernetes, Helm, ALB Ingress Controller, HPA, ACM, IAM OIDC*

Prometheus & Grafana Monitoring on Kubernetes [GitHub ↗](#)

- Deployed kube-prometheus-stack via Helm on EKS, collecting metrics from nodes, pods, and control-plane components.
- Built custom Grafana dashboards tracking pod CPU/memory, request latency, and node health with configurable alert thresholds.
- Configured Prometheus alerting rules and routed critical alerts to Slack via Alertmanager webhooks.

Tech: *Kubernetes (EKS), Helm, Prometheus, Grafana, Alertmanager, Slack*